Sanjana Venkatesh

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Education

B.E. Computer Science and Engineering | RNS Institute of Technology, Bengaluru | CGPA: 8.38/10 2023-2027

Experience

Software Developer Intern | IEEE TechXcelerate Jun 2025 - Oct 2025

- Engineered a physics problem simulator using Node.js and Express.js, enabling real-time 2D rigid body dynamics simulations with full-stack development approach.
- Implemented RAG (Retrieval-Augmented Generation) pipeline using Gemini-embedding-001 to parse complex user physics statements and map them to simulation using cosine similarity parameters achieving high accuracy.
- Designed and managed PostgreSQL schema to store and retrieve simulation states and user-generated problem templates.
- Integrated Matter.js and P5.js with RESTful API to render dynamic, parameter-driven visual simulations based on AI-parsed inputs.
- Architected and deployed a library of 39+ modular physics simulations (Kinematics, Electromagnetism, Dynamics) using P5.js and Matter.js, ensuring real-time rendering and physical accuracy for complex user-defined parameters.

Undergraduate Research Foundation (URF 2025) Apr 2025 - Dec 2025

- Received grant to pursue research on NLP, GGWave, Alexa skill development, and IoT for smart home automation for physically challenged individuals.
- Developing sign language translator using computer vision and communicate with Amazon Alexa device using GGWave technology for short distance data transmission.
- Conducting research and development of GGWave and its applications with NVIDIA Jetson Nano.

Projects

SimuPhysics: AI-Powered Physics Simulation Platform | Full Stack, AI Jun 2025 - Oct 2025

- Full-stack development of physics problem simulator using HTML, CSS, JavaScript, Matter.js, Google Gemini, REST API, Node.js, and Express.js with real-time 2D rigid body dynamics simulations.
- Implemented AI and RAG Architecture using Gemini-embedding-001 for parsing complex user physics statements and mapping them to simulation parameters with cosine similarity achieving high accuracy.
- Designed PostgreSQL database schema for simulation states and user-generated templates with frontend integration using Matter.js and P5.js for dynamic visual rendering.

Raspberry Pi GameGirl | Embedded Systems Jan 2025 - Mar 2025

- Engineered handheld console using RP2040 (Raspberry Pi Pico) microcontroller, implementing game logic and hardware abstraction layers in MicroPython.
- Optimized data transfer via SPI Protocol for 1.7" LCD (ST7735) and integrated high-capacity microSD module to manage file system for external game storage.
- Developed custom GPIO-based button matrix driver to handle real-time user input and interrupt-driven game state transitions.
- Designed expandable storage architecture supporting up to 8GB of external data, allowing dynamic loading of multiple retro game binaries.

Smart Home Automation for Physically Challenged | IoT, AI Apr 2025 - Dec 2025

- Research project on smart home automation using Alexa and NVIDIA Jetson Nano with focus on NLP, GGWave, Alexa skill development, and IoT integration.
- Developing sign language translator using computer vision and GGWave technology for short distance data transmission with Amazon Alexa device.

Leadership & Activities

Core Team Member, BigO Club (Competitive Coding Club): Hosted several events including Intro to Git and coding competitions for fellow peers. (Oct 2024 - Present)

Technical Skills

Languages: JavaScript (ES6+), Python, C, C++
Backend Development: Node.js, Express.js, FastAPI, Django, RESTful APIs
Frontend Development: React.js, P5.js, Matter.js, HTML5, CSS3
Databases: MySQL, PostgreSQL, Gemini Embeddings
Tools & Version Control: Git (CLI & GUI), GitHub, Docker
Cloud & Platforms: Linux (Ubuntu terminal), Windows, AWS (EC2/S3)
AI/ML: Generative AI (Gemini 2.5 Pro), RAG Architecture, NLP, Computer Vision
Embedded Systems: Raspberry Pi Pico (RP2040), MicroPython, SPI Protocol, GPIO

Certifications & Achievements

Samsung Innovation Campus - IoT: Selected among **800+ applicants**. Completed intensive training in Internet of Things, embedded systems, sensor integration, and IoT application development.

NPTEL Certifications: Earned two certifications in core computer science and engineering domains, demonstrating strong foundational knowledge.

Competitive Programming: Solved **150+ problems** on LeetCode, demonstrating strong proficiency in data structures and algorithms.

Minor Degree in VLSI: Currently pursuing minor degree in VLSI (Very Large Scale Integration), expanding expertise in semiconductor design and digital systems.